RESEARCH | Article in Press

Artificial intelligence in cellular senescence research: a systematic review assessing methodological quality and reporting standards using PROBAST + AI and TRIPOD + AI

Chanda Bhalchandra Rajurkar · Bala Murugan · Ganesh Pandian Namasivayam

Received: 22 March 2026 / Accepted: 8 June 2026
© The Author(s) 2026

Abstract

Cellular senescence is a fundamental mechanism of biological ageing that has emerged as a critical target for therapeutic intervention in age related diseases. The coalesce of artificial intelligence and senescence research provides unprecedented opportunities in advancing our knowledge and treatment approaches. This systematic review study addresses the gap across diverse AI model and the heterogeneity in senescence, by conducting the extensive evaluation of the performance outcomes and methodological rigor of AI models using Prediction model Risk of Bias Assessment Tool+Artificial Intelligence (PROBAST+AI) and reporting completeness using Transparent Reporting of a multivariable prediction model for Individual Prognosis or Diagnosis+Artificial Intelligence extension (TRIPOD+AI), providing the insights into AI models robustness and generalizability. For the articles, released between 2019 to 2025, across major databases 18 eligible studies was used for review, following PRISMA guideline. Quality and applicability were assessed by PROBAST+AI (4 domain) and reporting via TRIPOD+AI (27 items). The quantitative synthesis indicates that deep learning architectures, especially Convolutional Neural Networks (CNNs), are dominant which appeared in about 50% of the studies. These CNNs consistently outperform traditional machine learning methods in the analysis of morphological heterogeneity. While reported performance metrics were high, with accuracy ranging from 83.55% to 99.79%, the PROBAST+AI assessment indicates a high risk of bias in 83.33% (15/18) of studies, primarily driven by Analysis domain due to improper data splitting (data leakage) and lack of external validation. As well as adherence to TRIPOD+AI reporting standards was suboptimal with average of 62% 'YES'; notably, with the major gap in 0% of studies pre-registered a protocol and only 44.4% made analytical code publicly available, severely limiting reproducibility. Evidently AI demonstrates immense potential to accelerate biomarker discovery and senolytic drug screening, particularly through label-free morphological analysis by DL models, despite of high quality concern and poor reproducibility limit reliability; also standardization, shared benchmarks, multi-omics integration, and explainable AI are essential concerns for clinical translation in aging research.

Keywords Cellular Senescence · Artificial Intelligence · Systematic Review · TRIPOD+AI · PROBAST+AI

This Article in Press is shared early to give you faster access to new research. It is citable and carries a permanent DOI. The final edited version will replace it automatically.

Published online: 05 September 2026

Abbreviations

AI	Artificial Intelligence
AR	Average Recall
AUC	Area Under the Curve
CBAM-YOLO	Convolutional Block Attention Module—You Only Look Once
CNNs	Convolutional Neural Networks
DL	Deep Learning
GANs	Generative Adversarial Networks
hMSCs	Human mesenchymal stem cells
LASSO	Least Absolute Shrinkage and Selection Operator
LOOCV	Leave-One-Out Cross-Validation
LIME	Local Interpretable Model-agnostic Explanations
ML	Machine Learning
mAP	Mean average precision
PRISMA	Preferred Reporting Items for Systematic Reviews and Meta-Analyses
PROBAST+AI	Prediction model Risk of Bias Assessment Tool+Artificial Intelligence
ROC	Receiver Operating Characteristic
RF	Random Forest
RNNs	Recurrent Neural Networks
SVM	Support Vector Machines
SA-β-gal	Senescence-Associated β-galactosidase
SHAP	Shapley Additive Explanations
SASP	Senescence-Associated Secretory Phenotype
TL	Transfer learning
TRIPOD+AI	Transparent Reporting of a multivariable prediction model for Individual Prognosis or Diagnosis+Artificial Intelligence
XAI	Explainable AI

Introduction

The coalescing of artificial intelligence (AI) with cellular biology has more technically connected our understanding for cellular senescence. From philosophical inquiries to rigorous scientific investigation in study of aging and cellular senescence has progressed for culminating in significant discoveries that continue to broaden our understanding of cell aging processes. In the context of cell cycle, when cell cycle does not proliferate, due to any of the reasons such as DNA damage, oxidative stress, and telomere attrition, this state of cell cycle arrest is cellular senescence state [1]. Studying this cell cycle state plays a crucial role in aging and age-related diseases [2]. Understanding senescent cells thus have become a prime point in aging research, with the understanding to bring advancements in therapeutic interventions for aging and age-related diseases [3].

With all the advancements in biology, artificial intelligence has also become transformative powerful tool in biomedical research. Artificial intelligence comes with the computational techniques including deep learning (DL) and machine learning (ML), that could analysis the complex datasets, can do pattern recognition, and could also do predictive modeling. With the reference to cellular senescence research, AI offers unprecedented opportunities to enhance the understanding of cellular aging by automating the detection of senescent cells, as well identifying novel biomarkers, drug discovery, and uncovering molecular mechanisms underlying senescence [4]. Current challenges in senescent cell research could be briefed as identification and characterization of senescent cells which was been historically limited research progress due to the several technical challenges. One of the

significant issues is heterogeneity, as senescent cell has diverse morphological and molecular signatures, making standardized detection difficult [5, 6]. Not only this but the complex phenotypes associated with senescence, particularly the senescence-associated secretory phenotype (SASP), could vary across different cell types and contexts [7]. The challenges increase with the traditional detection methods, as they could be very time consuming, laborious and often subjective cognition. As well to rely on identification of senescent cells there is need for multiple biomarkers and assays [8].

With all the advancements in biology, AI's computational techniques of ML and DL has also become a standard tool in biomedical research. These DL and ML, can analyze complex datasets, perform pattern recognition, and build predictive models. Beyond foundational biology, AI has recently established robust state-of-the-art benchmarks in digital medical imaging, demonstrating exceptional performance in complex diagnostic tasks such as the automated classification of diabetic macular edema and soft-computing-based brain tumor segmentation [9, 10]. Building upon these successes in broader pathology, in the reference to cellular senescence research, AI offers unprecedented opportunities to enhance the understanding of cellular aging by automating the detection of senescent cells.

Artificial intelligence has potential to address the above challenges, by offering the solutions to the limitation of identifying and characterizing senescent cells, as AI could automate the image analysis, deep learning algorithms could process large-scale microscopy data [11]. Even machine learning models excel at recognizing subtle phenotypic signatures that might have been overlooked by traditional methods [12]. AI is also capable of building multi-modal integration, enabling the multi meta data from various experimental techniques [13]. With the multi-AI model, the AI's predictive model building enables the prediction of senescence trajectories and therapeutic responses. This is the most powerful applications of AI, the ability to create predictive models. In the context of cellular senescence, AI can be used to predict how cells will behave over time, including their senescence trajectories and responses to various therapeutic interventions [14]. AI has much more potential to overcome several challenges in senescence research, like addressing the heterogeneity of senescent cell phenotypes, the complexity of SASP, and providing the high-throughput and quick analysis of cellular and molecular data.

For instance, in the study by authors (Maxwell Mai et al., 2023), they have trained a morphology-based deep learning model to classify human mesenchymal stem cells (hMSCs) based on differentiating categories adipogenic or osteogenic lineages, the high-content imaging data was given as input to the model to extract relevant morphological features that correlated with differentiation outcomes [15]. Similar way, the study by authors (Nagao et al., 2020) built deep learning model using convolutional neural networks (CNNs) applied to fluorescence microscopy images, to classify cell cycle phases [16]. The model input data is not only limited to the microscopic images, but AI is capable to analyze transcriptome data as well, for instance the study by (Lee et al., 2025), identified the best gene sets from transcriptome data for the classification of aging and senescence, they have also developments a AI tools like T-CLASS, which is a web-based platform that has shown robust and reliable results in a variety of experimental settings [17]. As well in the study by (Mahmud et al. 2024) used machine learning techniques, such as random forests, decision trees, and support vector machines while choosing a refined feature selection, to use transcriptomic signatures for distinguishing between senescent and healthy cells, to identify a small subset of genes that could be essential for precise classification [18]. With these above mentioned articles we could see the transformative potential of AI in accelerating the discovery of novel biomarkers, therapeutic targets, and interventions for age-related diseases in the next wave of AI.

With the understanding that AI has widen up the range in varying cellular senescence research, a need of systematic review and quality assessment analysis was essential to consolidate existing knowledge to evaluate the effectiveness of AI approaches, and to know when these coalescing domains comes to gather what next challenges occurs and to provide a systematic approach for future studies with AI governance. Moving beyond a traditional meta-analysis of performance metrics, this study also conducts a rigorous methodological audit using the newly established 2024/2025 extensions of the Prediction model Risk of Bias Assessment Tool+Artificial Intelligence (PROBAST+AI) and Transparent Reporting of a multivariable prediction model for Individual Prognosis or Diagnosis+Artificial Intelligence extension (TRIPOD+AI) frameworks. By applying these tools, we evaluate

the risk of bias, clinical applicability, and reporting transparency of AI models including machine learning (ML), deep learning (DL), and hybrid approaches used in detecting and characterizing senescent cells. The study also includes senescence cell detection and characterization by the evaluation of AI models performance across variety of dataset and in different experimental conditions which will help us to understand the impact of AI on senescence biology, which includes biomarker discovery and therapeutic target identification. The approach to utilize AI in cellular Senescence research has been put before by this review, which will coalesce interdisciplinary collaboration and accelerate the development of innovative solutions for classification of senescence cell which further aids the study of age related diseases. A comprehensive overview of current state of the field has been provided by this study which will serve as a valuable resource for practitioners and researchers that can be utilised in senescence cell research through AI. A comprehensive picture of the present state of AI in studying cellular senescence can be obtain from this study.

Methodology

Search strategy and selection criteria

The literature search was conducted on various online databases such as PubMed, Web of Science, ScienceDirect, IEEE Xplore, Springer, and Google Scholar, which were accessed via an institutional gateway. This access enabled the retrieval of high-impact, full-text records for eligibility assessment. The search carried on articles released in the recent seven years, from January 2019 to November 2025, and utilised the following primary search string: ("cellular senescence" OR "senescent cells" OR "senescence") AND ("artificial intelligence" OR "machine learning" OR "deep learning" OR "neural networks"), which was adapted to match the specific syntax, field tags, and Boolean structures of each individual database (see Supplementary Table S1). The primary search filtered for the terms anywhere in the article, not just in the title, keywords, or abstract. The advanced search criteria included filtering for journal articles, English language publications, and peer-reviewed studies or established preprint repositories such as bioRxiv. The searched terms on databases were "detection", "analysis", "identification", and "classification". The inclusion criteria were original research articles published between 2019 and 2025, studies applying AI/ML/DL methods to senescent cell research with the specified analysis terms, and studies with quantifiable outcomes. Exclusion criteria included articles like review articles without original data, conference papers, studies without clear AI methodology and non-English publications.

PRISMA flow diagram

For systematic review, study selection process was conducted in accordance with the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) guidelines [19, 20]. By conducting a PRISMA study enhanced the transparency, methodological quality, and reproducibility of systematic review, ensuring comprehensive reporting and reducing bias in research [19, 21]. The PRISMA flow diagram in Fig. 1 illustrates the various stages of the systematic review like identification, screening, eligibility, and inclusion [22–24].

Identification

The database search results showed total of 6,415 records at primary search across six major electronic sources, given as PubMed ($n=565$), ScienceDirect ($n=417$), Web of Science ($n=65$), Springer ($n=115$), IEEE Xplore ($n=153$), and Google Scholar ($n=5,100$). After applying the advanced search criteria included database-level filters as mentioned in Sect. "Search Strategy and Selection Criteria" used to exclude ($n=2,181$) article. The remaining entries were then imported into Mendeley Reference Manager, where ($n=3,785$) duplicate records were identified and removed. Additionally, 28 records were manually removed for reasons like publication type

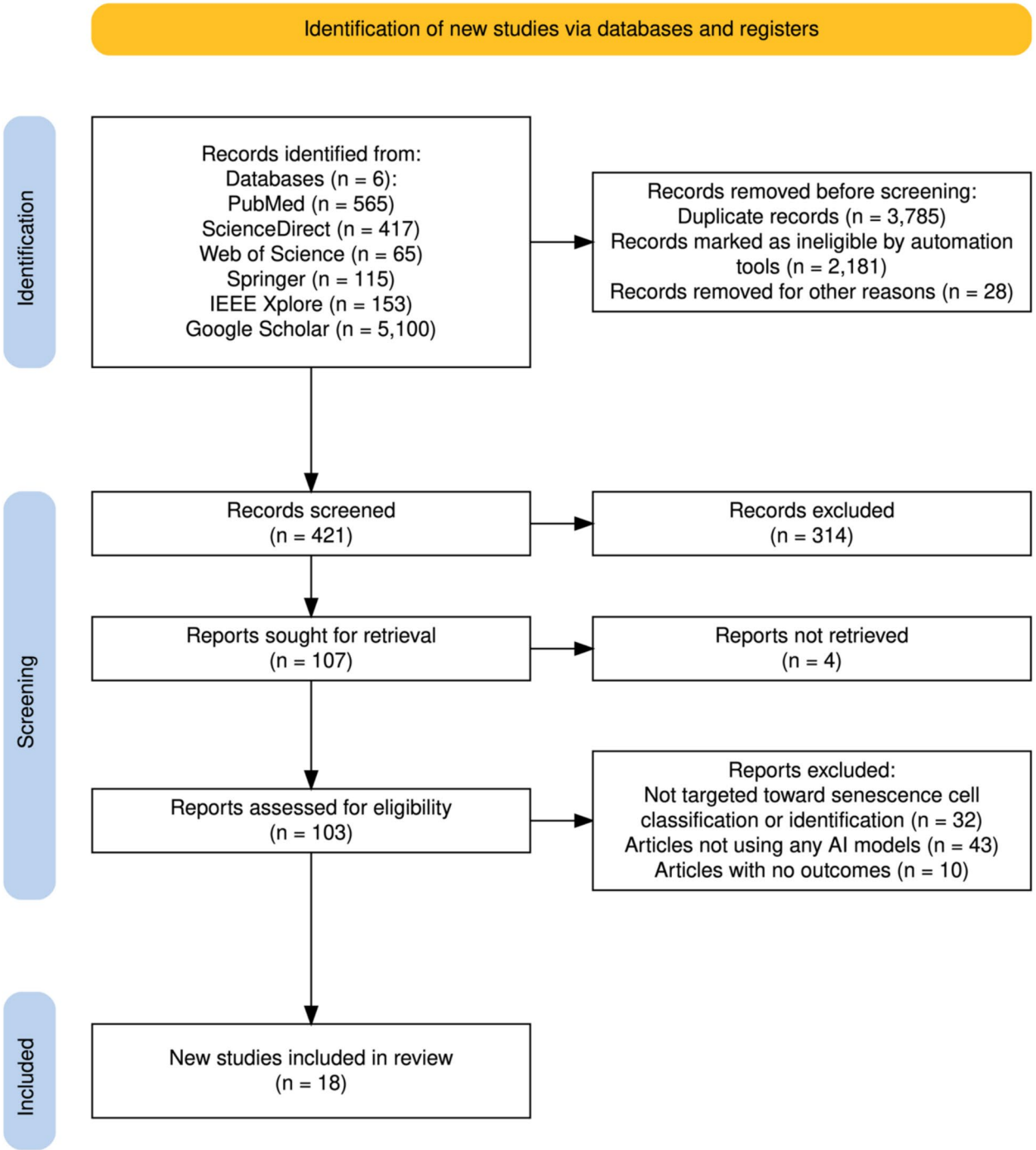


Fig. 1 PRISMA flow diagram. The PRISMA flow diagram was created using shinyapps web app in reference to article "PRISMA2020: An R package and Shiny app for producing PRISMA 2020-compliant flow diagrams, with interactivity for optimised digital transparency and Open Synthesis" by Haddaway et al., 2022 [22, 23]

(e.g., editorials, patents) or non-accessible content. After this filtration step, 421 records were retained for detailed title and abstract screening.

Screening and eligibility

Based on title and abstract relevance, 421 records were screened by a standardised two-stage screening process that was executed to minimise human bias. In the first stage, two reviewers (C.R. and B.R.) independently evaluated the titles and abstracts of the 421 records against the core inclusion and exclusion criteria. A primary biological cause for exclusion at this juncture was studies focusing on plant senescence (e.g., crop or leaf ageing), which did not align with the biomedical and preclinical scope of this review. This initial screen resulted in the exclusion of 314 records irrelevant to senescent cell research or AI methodology, leaving 107 full-text articles to be sought for retrieval. Four full-text articles could not be accessed due to restricted availability or download failure. The remaining 103 full-text articles were independently evaluated in detail by both reviewers. Based on predefined criteria, 83 articles were excluded for the following reasons:

- Studies not directly addressing senescent cell classification or identification ($n=32$)
- Absence of AI, machine learning, or deep learning methodologies ($n=43$)
- Lack of quantifiable outcomes relevant to model performance ($n=10$)

Any scoring or eligibility discrepancies between the two primary reviewers during either stage were resolved through collaborative discussion and line-by-line consensus. If an agreement could not be reached, a third reviewer (G.P.) was available for final arbitration.

Inclusion

After applying all inclusion and exclusion criteria, 18 studies were determined to be eligible and were included in the final systematic review and further PROBAST+AI and TRIPOD+AI study analysis.

Rationale for framework selection PROBAST+AI and TRIPOD+AI were applied only to the primary included studies that developed or evaluated AI-based predictive or classification models in cellular senescence. Supporting references used for background, methodological context, or discussion were not assessed with these tools. While PROBAST+AI and TRIPOD+AI were originally designed for clinical prediction models, their application to the biological studies included in this review (References [16–32] and [33]) is justified by the functional nature of the AI tasks described.

Diagnostic Equivalence: The AI models utilized in the primary studies function as multivariable diagnostic tools, classifying cellular states (senescent vs. non-senescent) based on morphological, proteomic, or transcriptomic features.

Prognostic Application: Several included studies use AI to predict treatment responses or "senescence trajectories," which aligns with the prognostic focus of the PROBAST framework.

Methodological Rigor: Applying these high-level clinical standards allows for a rigorous audit of common AI pitfalls in biology, such as data leakage (splitting by cell vs. by replicate) and lack of external validation, which are critical for the eventual translation of these findings into clinical aging research.

To guarantee the objectivity and scientific integrity of the methodological audit, the PROBAST+AI and TRIPOD+AI assessments were executed via a rigorous, independent dual-reviewer protocol. Two reviewers (C.R. and B.R.) applied the frameworks independently to all 18 included studies. For the quality and applicability audits, each of the 4 domains in PROBAST+AI was evaluated by systematically answering its specific AI-extended signalling questions to arrive at a conservative overall risk judgement. For reporting quality, the

reviewers independently scored the 27 selected checklist items from TRIPOD+AI as 'Yes' (fully reported), 'No' (not reported), 'Partial', or 'N/A'.

Following this independent coding phase, both reviewers compiled and cross-examined their respective evaluation sheets. While a formal inter-reviewer Cohen's kappa statistic was not calculated, a strict line-by-line consensus protocol was enforced. Discrepancies primarily occurred during the assessment of the PROBAST+AI Analysis domain (regarding the exact boundary definitions of data leakage) and TRIPOD+AI sample size justifications. These minor scoring variances were systematically debated and resolved through collaborative re-review of the primary source texts until a 100% mutual consensus was achieved across all domains and items, eliminating the need for third-party arbitration.

Quality assessment using PROBAST + AI

The quality of the included studies was assessed using the PROBAST+AI (Prediction model Risk of Bias Assessment Tool+Artificial Intelligence) tool [25]. This tool evaluates the risk of bias across 4 domains: Participants, Predictors, Outcome and Analysis [20, 34].

The PROBAST+AI assessment each domain was rated either of low, high, or unclear concern based on the signalling questions, with an overall judgement derived conservatively (high if any domain was high) [34]. The results of the quality concern assessment across all 4 domains with overall ratings are highlighted in Fig. 3 (PROBAST+AI quality concern ratings by domain and overall) and summary in Fig. 4 (Summary of PROBAST+AI quality concern across domain and overall). As well the applicability concerns where evaluated for the 3 domains (Participants/data sources, Predictors and Outcome) with an overall applicability ratings, relative to the intended use of preclinical or research cellular senescence classification which is highlighted in Fig. 5 (PROBAST+AI applicability concern ratings by domain and overall) and summary in Fig. 6 (Summary of PROBAST+AI applicability concern across domain and overall). The complete, itemized signalling-question checklist scoring matrix for each individual study is provided in supplementary table S2 and supplementary table S3 to ensure full audit access. The charts were prepared using robvis, a web app designed for visualizing domain assessments performed as part of a systematic review [26, 27].

Reporting quality assessment TRIPOD + AI

We utilized TRIPOD+AI (Transparent Reporting of a multivariable prediction model for Individual Prognosis or Diagnosis – Artificial Intelligence extension) checklist to evaluate the transparency and completeness of each included study of reporting the key aspects of their prediction model [28, 29]. Due to the large number of studies and the focus on AI-specific reporting gaps in senescence classification models, we assessed adherence to 27 items from checklist (in total of 53 including the sub-items). Every item for the included articles was scored as "Yes" (fully reported), "No" (not reported), "Partial" (partially reported), or "Not Applicable (N/A)". The adherence rate for each study was subsequently determined by dividing the total number of applicable items by the number of "Yes" scores; see Table 2. The purpose of this analysis was to find reporting standard gaps, particularly with regard to model architecture, code sharing, and validation processes.

Result

Systematic analysis result

The application of AI-driven methods for cellular senescence detection and classification is examined in this systematic review study. According to the analysis of the reviewed articles, morphological quantification and deep learning algorithms are essential for improving the accuracy and scalability of senescence detection in various

cell types and biological contexts. The main themes and takeaways from the reviewed studies are highlighted in the discussion that follows. The review's findings are further detailed in Table 1, which provides a comprehensive summary of the included articles.

AI models and techniques used for senescence cell classification

In the included articles, the most commonly used AI models for cellular senescence study were deep learning techniques, such as recurrent neural networks (RNNs) and convolutional neural networks (CNNs), which are often applied to image-based classification (e.g., microscopy images, cell morphology) for morphological features extraction, relatively approximate 50% of included study used Deep learning algorithms [30, 31]. Like, the article by authors He, Liangge et al., 2024 used CNNs to analyse cell morphology and identify senescence-related changes such as nuclear enlargement and altered cell shape [36]. And in study by authors Kusumoto, Dai et al., 2021, they choose a deep learning approach to assess the effects of potential anti-senescent compounds on cell morphology [30]. For high-content imaging, the authors Wang et al., 2025, implemented deep learning pipelines to map and objectively quantify senescent stromal cells, which enhanced scalability and reduced human bias [46]. Along with DL, often ML models like support vector machines and random forests is also built in studies, relatively approximate 30% of included study used traditional machine learning algorithms mostly for classification tasks involving high-dimensional feature spaces derived from microscopy images or gene expression dataset [12, 41].

Along with the deep learning and machine learning techniques, many articles have used ensemble learning and transfer learning, for example the article by authors [38], these techniques combine multiple AI models to improve the robustness and generalizability of the classification process, especially when we are working with small or noisy datasets. Transfer learning models, which are utilized when datasets are small and specific models are trained on pre-trained networks (e.g., using ImageNet or other large datasets for fine-tuning), relatively approximate 10% of included study used transfer learning [38]. Ensemble learning methods were also used combining multiple models for more robust predictions, relatively approximate 10% of included study used Ensemble learning techniques [32, 41].

Biomarkers associated with cellular senescence for included studies

Traditionally, the identification of senescent cells is linked with certain biomarkers for study. Several biomarkers are used for identifying senescent cells, particularly in combination with AI models. The most commonly referenced biomarkers used in included articles was cellular morphology, appearing in seven studies (38% of total included articles). It reflects its widespread adoption as a key feature for identifying senescence-related changes, particularly in morphological deep learning frameworks. Like For example, nuclear abnormalities such as enlarged nuclei, heterochromatin foci, and nuclear irregularity are central features used for classification [12]. Nuclear morphology is used in 4 studies (22% of total included studies), it is often used as a direct biomarker for identifying senescent cells, emphasizing its relevance in senescence-specific changes, as seen in articles investigating nuclear features [35]. SA-β-gal (Senescence-Associated β-Galactosidase), a widely used biomarker for detecting senescence, p16INK4a, p21 which are key regulators of cell cycle arrest in senescence, p53 a tumor suppressor gene often implicated in the initiation of senescence; these proteomics as biomarkers used in 5 studies (28% of total inclusive articles). Transcriptomics leveraged RNA sequencing used in 3 articles (17% of total inclusive studies), to classify cells based on gene expression patterns, this technique underscores the growing integration of molecular-level data into machine learning models. While in the study by Wang et al. (2025), they came across highly effective strategy for identifying senescent cells by integrating cell and nuclear morphological features with molecular biomarker [46]. In which Wang et al., 2025 utilized deep learning techniques on high-content microscopy images of mesenchymal stromal cells, where they have systematically evaluated morphological features such as cellular outline, nuclear dimensions, aspect ratio, and convexity., this approach

Table 1 Included article analysis

Study Reference	Paradigm	Architecture/Algorithm Used	Senescence Biomarkers Targeted	Primary Dataset Source	Target Computational Outcomes	Extracted Performance Parameters	Reported Performance Metrics/Scores
Heckenbach,Indra et al., 2022 [35]	DL	CNN	Nuclear morphology	Custom microscopy images	Senescent Cell Classification	Accuracy	Accuracy: 95.00%
Kusumoto, Dai et al., 2021 [30]	DL	Transfer Learning (CNN)	Cell morphology	Phase-contrast microscopy images	Anti-Senescent Drug Efficacy Scoring	Accuracy	Accuracy: 90.00%
He, Liangge et al., 2024 [36]	DL	RCNN	Cell morphology	Proprietary MSC Microscopy Images	Automated Senescence detection	Mean Average Precision (mAP), Average Recall (AR), Precision, Recall, F1-Score	mAP: 0.81, AR: 0.93, Precision: 0.85, Recall: 0.923, F1: 0.885
Weber, Leya et al., 2023 [37]	DL	GANs	SA-β-gal, Protein expression	Proprietary microscopy images	Phenotypic scoring	-	Not Reported
Sorrentino, Salvatore et al., 2023 [38]	DL	Hybrid Architecture (Ensemble+TL/ML)	Nuclear features	Public nonlinear optical microscopy	Senescent cell classification	Accuracy, Precision, Recall, Recall F1-Score, AUC	Accuracy: 90.10%, Precision: 95.70%, Recall: 87.40%, F1: 90.80%, AUC: 0.96
He, Sarah et al.,2024 [32]	DL	DL (Hybrid)	Cellular Morphology	Synthetic Label-Free Images	Multi-State Live-Cell Tracking	Accuracy	Accuracy: 96.40%
Jiang, Hanli et al.,2024 [31]	DL	RNN	Protein sequences	Public repositories (proteomic)	Senescence classification	Accuracy	Accuracy: 83.55%
Heckenbach, Indra et al., 2021 [39]	DL	Hybrid DL (Xception Core)	Cellular markers	Public microscopy images	Prognostic analysis	Accuracy	Accuracy: 95%
Yang, Ruixin et al., 2023 [33]	DL	CBAM-YOLOv3, YOLOv3	SA-β-gal Staining	Organoid Microscopy Images	Scoring senescence	mAP, F1-score, Recall, Precision	mAP: 93.20%, F1: 0.86, Recall: 82.88%, Precision: 88.43%
Aspert, Théo et al.,2022 [40]	DL	CNN	Cell division markers	Custom Time-Lapse Imaging	Division survival analysis	Precision	Precision: 90.00% – 95.00%
Duran, Imanol et al., 2024 [41]	ML-SVM	Decision Tree Algorithms (5 models), Random Forest Algorithms (3 models), Consensus Algorithm RF	Nuclear morphology features	Microscopy image Repositories	Senescent Cell Classification	Accuracy, Precision, Recall, F1-Score	Accuracy: 89.00%, Precision: 90.00%, Recall: 94.00%, F1: 89.00%
Malavolta, Marco et al., 2022 [42]	ML	Random forest	Nuclear features	Imaging flow cytometry	Unstained Live-Cell Detection	Sensitivity, Specificity	Unclear Performance Parameters

Table 1 (continued)

Study Reference	Paradigm	Architecture/Algorithm Used	Senescence Biomarkers Targeted	Primary Dataset Source	Target Computational Outcomes	Extracted Performance Parameters	Reported Performance Metrics/Scores
Jochems, Fleur et al., 2021 [43]	ML	Ensemble learning	Senescence markers in cancer pathways-Proteomics	Cancer cell line Proteomics	Comprehensive Delineation Profiling	Accuracy, Precision, Sensitivity, Specificity	Accuracy: 90.00%, Precision: 0.94, Sensitivity: 0.80, Specificity: 0.97
Hughes, Bethany K. et al., 2023 [44]	ML	Random forest, logistic regression, Multiple Discriminant Analysis(MDA)	Transcriptomic features (senescence-associated gene expression)	Single-cell RNA sequencing data	In Vivo Senescent Burden Classification	Accuracy	Accuracy: 99.79% (True Positive Rate)
Yang, Ruida et al., 2024 [45]	ML	LASSO regression, RF, Multi algorithms(113) using LOOCV framework	Transcriptomic and clinical features, Molecular biomarkers (senescence gene expression)	Transcriptomic datasets and patient data (RNA-seq data from intrahepatic cholangiocarcinoma patients)	Predictive/prognostic senescence model	AUC	AUC: 0.72925
Welter, Emma M. et al., 2024 [12]	Hybrid DL+ML	Random forest, CellProfiler, Ilastik: This software uses random forests, a supervised machine learning algorithm, Logistic regression—Primary algorithm	Morphological changes in fibroblasts (size, flattening, granularity)	Microscopy images of replicative senescence in human fibroblasts	Replicative Senescence Identification	ROC AUC, AUC, Classification accuracy (CA),F1, Precision, Recall	Accuracy: 0.68, Precision: 0.681, Recall: 0.681, F1: 0.676, ROC AUC: 0.84, AUC: 0.741
Wang, Liudi, et al., 2025 [46]	DL	ResNeXt Architecture (CNN)	morphological changes, Ki67 (specifically Ki67(−)), Phosphorylated ribosomal protein S6 (pRPS6) (specifically pRPS6(+)), SA-β-gal (specifically SA-β-gal(+)), P16, P21 and p53	Optical microscopy images of MSCs at different passages (cell generations)	Predicted generation number indicating cell senescence stage, and classification of MSC type (umbilical cord vs bone marrow)	Accuracy	Accuracy: 90.2%
Mahmud, Shamsed, et al., 2024 [18]	ML	SVM, RF, Decision Tree, and Soft Independent Modelling of Class Analogy (SIMCA) or RSimca	RNA sequencing	Transcriptome data	Identifying and classifying cellular senescence from bulk RNA sequencing data	Accuracy	Accuracy: 99.50%

was supported by validating with molecular markers like loss of Ki67 (specifically Ki67-negative) expression, activation of pRPS6, elevated SA-β-gal, and changes in P16, P21, and p53 levels [46]. By combining both the approach of morphological and molecular information, the authors methodology supports precise and scalable enumeration of senescent populations [46]. SA-β-gal staining, widely recognised as a hallmark of senescence, is frequently used in conjunction with AI-based methodologies to determine the presence of senescence [33, 37]. Some studies also asses other biomarkers such as p16INK4a, p21, and p5, to establish a correlation between gene expression with morphological features [31, 37]. Consequently, while morphological features remain the most common form of data input, the integration of molecular biomarkers into AI-driven models is facilitating a more comprehensive and nuanced evaluation of senescence phenotype.

Dataset diversity and methodological breadth

The datasets used across the studies vary widely, with some relying on microscopy images which were the most commonly used, appearing in 12 out of 18 studies (67%), and 2 out of 12 were proprietary datasets [36, 37]. Synthetic dataset was also observed in study (5% out of inclusive articles) generated through simulation tools or augmentation techniques to enrich the training set and avoid overfitting [32]. The use of public cancer datasets was also observed in article, such as by authors Yang, Ruida et al., 2024 used The Cancer Genome Atlas (TCGA), which was used for predictive models that assess senescence burden in cancer cells and their potential relationship with prognosis [45]. Thus, incorporating clinical datasets demonstrates the utility of AI in clinical decision-making. Other specialized imaging techniques, such as imaging flow cytometry, were also used in study by Malavolta, Marco et al., 2022, accounting for 5% of the total inclusive articles [42]. Molecular datasets, including protein sequences repositories, cancer cell line proteomic datasets, and single-cell RNA sequencing, were used in a total of 3 studies (15%), these datasets demonstrate an emerging trend in integrating omics-level data to complement traditional imaging techniques [31, 43, 44]. In one of the studies by Aspert, Théo et al., 2022, they used custom data for Time-Lapse Imaging [40]. With the recent developments in dataset collections, one study used transcriptome-based AI platforms named as T-CLASS, which enables effective gene signature extraction but depending on benchmark datasets for reliable validation in newer biological systems [17]. The diversity in dataset reflects the ongoing innovation in senescence research with an increasing focus on combining imaging and molecular data for more comprehensive analyses. However, the predominance of microscopy datasets underscores their central role in senescence detection and classification workflows.

AI model output and prognostic insights

After analyzing the AI models outcomes, the most common model output among the studies was cell classification, reported in 8 out of 18 articles (44%). This reflects the central role of classification models in cellular senescence. Some 3 studies (17%) highlight the scoring as output outcomes of the AI models for generating senescence scores for drug efficacy scoring, phenotypic scoring, senescence scoring for predicting efficiency of treatments that target senescence in gastric organoids [30, 33, 37]. Few articles (11% out of total inclusive articles) also describe outcomes to detect the senescence, for cellular aging in regenerative medicine, and in another to detect the live senescence [36, 42]. In 2 of the studies (11% out of total inclusive articles) the model outcomes for the prognostic analysis, how senescence can act as a barrier to cancer development by suppressing cell proliferation and promoting cell death or cycle arrest [39], another study model prognostic outcomes for survival predictions or the likelihood of treatment resistance, based on the senescence burden in the tumor [45]. The AI model in 1 study (5%) is also used for the outcome of cell division tracking identifying the start and end of division events [40]. One another study by Jochems, Fleur et al., 2021 model outcomes predicting comprehensive profiling of senescence [43]. Report of model in 1 study (5%), identifies replicative senescence focus on quantifying numerical characterization of senescence at the cellular level. Another study by Wang et al., 2025 reported that their model could predict the generation number of mesenchymal stromal cells, effectively indicate the senescence

stage, while also enabling classification between MSC types derived from umbilical cord and bone marrow sources [46].

Analysis parameters and performance metrics

Across all the included articles to evaluate the performance of AI models various metrics were used in, including sensitivity, accuracy, F1-score, specificity, precision, AUC (area under the receiver operating characteristic curve), and mean average precision (mAP). As well, the cross-validation techniques have also frequently been used to assess model robustness [39]. Few studies also explore model explainability, such as feature importance in random Forests, and the use of gradients or attention maps for deep learning models to interpret which features contribute most to senescence classification (Sorrentino, Salvatore et al., 2023) [38]. Accuracy was the most frequently reported metric, appearing in 14 of the 18 analysed studies (78%), reflecting its central role as a benchmark for determining model effectiveness in senescence detection tasks, refer Fig. 2a. However, a critical audit reveals profound inconsistencies in parameter reporting across the literature. Cellular senescence datasets are frequently plagued by severe biological class imbalances, where senescent cells represent a small minority of the total population. In these contexts, relying exclusively on raw accuracy scores can be highly deceptive, potentially masking an algorithm's inability to reliably detect true positive events.

Despite this mathematical risk, robust metrics designed to evaluate class-imbalanced performance were severely underreported. The metrics of precision (utilized in 39% of cases) and recall (applied in 28%) were used irregularly to evaluate the correct identification of true positives (refer Fig. 2b and d). The AUC score, reported in only 3 studies (17%), was highly valued for its ability to measure performance over varying decision thresholds (refer Fig. 2e), while the F1-score was extracted from 5 studies (28%) to provide a balanced evaluation of precision and recall, particularly in condition of imbalanced datasets (refer Fig. 2c). Also 2 studies (11% of studies)

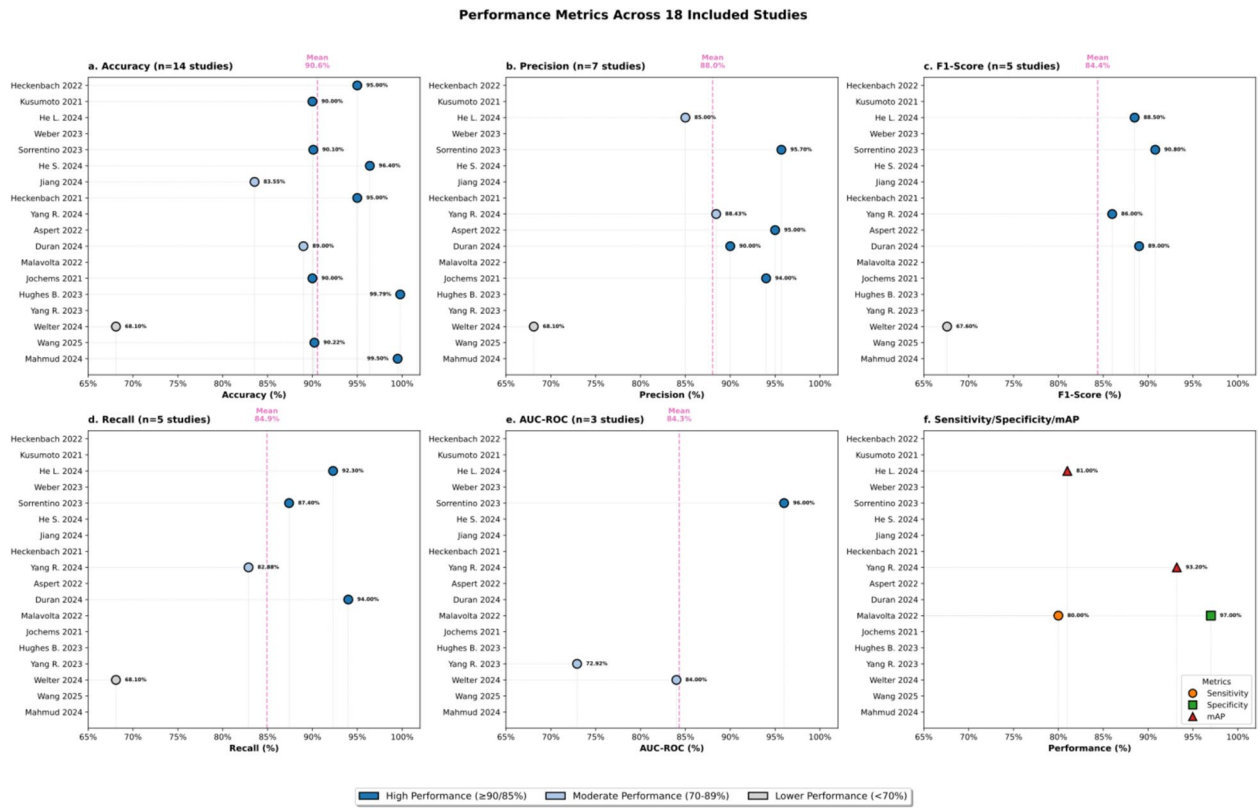


Fig. 2 AI model performance metrics across included studies

reported mAP (mean Average Precision), refer Fig. 2f. As well the author Malavolta, Marco et al., 2022 reported specificity and sensitivity, refer Fig. 2f [42]. Crucially, clinical standards like sensitivity and specificity were rarely co-reported. This fragmented reporting landscape underscores an urgent requirement for future studies to abandon ad-hoc metric selection and strictly align their evaluation methods with the standardized reporting guidelines set forth by the TRIPOD+AI framework.

Overall findings from all-inclusive study demonstrates superior accuracy of AI in senescent cell identification compared to traditional methods. Deep learning models have demonstrated a promising capacity for the morphological analysis of senescence cells; moreover, the integration of diverse data modalities has been shown to significantly enhance predictive accuracy.

Synthesis of technological and methodological trends

The integration of AI into cellular senescence research reveals a significant technological divide governed by the specific scale of biological data being analysed. A cross-analysis of the included studies indicates that image-based research, comprising 67% of the reviewed literature, has undergone a definitive transition from traditional machine learning to deep learning (DL) architectures. For instance, studies by Heckenbach, Indra et al., 2022 [35] Kusumoto, Dai et al., 2021 [30] built Convolutional Neural Networks (CNNs) to automate the detection of morphological hallmarks, such as nuclear enlargement and increased granularity, which are often too subtle for manual quantification. Conversely, research utilizing transcriptomic or proteomic data, such as Hughes, Bethany K. et al., 2023 [44] and Mahmud, Shamsed, et al.,2024 [18], maintains a reliance on traditional Machine Learning (ML) models like Random Forests and Support Vector Machines. This preference is not due to a lack of computational power but rather a prioritization of biological interpretability; these models allow for the extraction of "feature importance" scores, which are essential for identifying the specific gene sets or protein sequences that drive the senescent phenotype.

A critical finding emerging from this synthesis is the "Performance-Validation Paradox," where reported model accuracy appears inversely correlated with methodological rigor. While studies like Mahmud, Shamsed, et al., 2024 [18] and Hughes, Bethany K. et al., 2023 [44] report exceptional performance metrics, including accuracies of 99.50% and 99.79% respectively, these same studies were flagged with a "High Risk of Bias" in the Analysis domain of our PROBAST+AI audit. This discrepancy highlights a pervasive issue of data leakage and a lack of external validation within the field. Many high-performing models were evaluated on cell populations from the same biological batch or culture dish used for training, meaning the AI may have learned to recognize specific laboratory conditions (batch effects) rather than generalizable biological features of senescence.

Finally, the data suggests an ongoing evolution in the very definition of a senescence biomarker, moving from static, single-stain assays toward complex, multi-modal computational signatures. The approach by Wang, Liudi, et al., 2025 exemplifies this trend by systematically integrating classical molecular markers, such as SA-β-gal and p16, with high-content deep learning features derived from mesenchymal stromal cells. This shift indicates that the future of the field lies in hybrid methodologies that coalesce structural morphological data with functional omics data. By identifying these trends, it becomes clear that while AI has reached a state of high performance in senescence detection, the next phase of research must prioritize standardization and the elimination of systematic biases to achieve clinical reliability.

Critical methodological audit (PROBAST + AI) and reporting standards (TRIPOD + AI)

A rigorous methodological audit was performed utilising two complementary frameworks specifically designed for artificial intelligence-based prediction models: PROBAST+AI was employed to assess potential biases and applicability concerns, and TRIPOD+AI was utilised to evaluate the transparency and completeness of reporting [28, 34]. The PROBAST+AI tool systematically examined internal validity across four critical domains that are Participants or Data sources, Predictors, Outcome, and Analysis, in addition to assessing applicability within the

context of preclinical cellular senescence classification [47]. Concurrently, the TRIPOD + AI checklist scrutinized adherence to recommended reporting standards against 27 key items, which is essential for ensuring reproducibility and maintaining scientific integrity. Collectively, these assessments offer a critical appraisal of the methodological quality, generalizability, and transparency of AI applications in the detection and characterization of senescent cells, thereby highlighting both the current strengths and the pervasive limitations within the existing body of evidence.

Quality concern assessment (PROBAST + AI)

The application of PROBAST + AI in this review article will reveal a pervasive quality concern across the included studies. The PROBAST + AI quality concern assessment ratings are shown in Fig. 3 and further summarised in Fig. 4, while this reporting demonstrates that 15 out of 18 evaluated studies (83.33%) were rated as High Risk of Bias overall. The primary driver was the Analysis domain (domain 4). This is often due to improper handling of data splitting (data leakage), where the independence of the training and validation sets could not be guaranteed, or failure to account for overfitting in small sample sizes. Many studies split data by cell rather than by biological replicate or patient. If cells from the same culture dish are split into training and testing, the AI learns batch effects (e.g., lighting conditions of that specific dish) rather than biological features of senescence. Only He et al. (2024), Yang et al. (2023), and Duran et al. (2024) demonstrated low risk in the analysis domain.

The pervasive high risk of bias identified in 83.33% of the included studies; refer Fig. 3, is not merely a statistical observation but a reflection of systemic methodological vulnerabilities in bio-imaging AI. The primary driver, Domain 4 (Analysis), revealed a frequent failure to maintain 'biological independence' during data splitting. Specifically, several studies performed splits at the cellular level rather than by biological replicate or patient source. This creates a high risk of data leakage, where the model essentially 'memorizes' the batch effects such as the specific microscopy settings or culture conditions of a single dish rather than identifying generalizable morphological features of senescence. We observe the contrast between high overall risk (83.33%) and high applicability (94.44%) that reveals a critical 'Translational Gap'. While the models are designed to address highly relevant preclinical questions such as drug efficacy and biomarker identification for which the internal validity is compromised by a lack of external validation and insufficient accounting for model overfitting. This suggests that while the field is targeting the correct biological outcomes, the current analytical pipelines lack the rigor required for regulatory or clinical acceptance.

Domain 1 (Participants or Data Sources) was rated low in 10 studies (55.55%), high in 6 (33.33%), and unclear in 2 (11.11%), primarily due to small or proprietary datasets. Domain 2 (Predictors) was predominantly low across all 18 studies (100%), which reflects appropriate use of senescence-relevant biomarkers (e.g., nuclear morphology, SA-β-gal). Domain 3 (Outcome) showed low concern in 15 studies (83.33%), high in 2 (11.11%), and unclear in 1 (5.55%), with occasional gaps in ground truth reliability. Domain 4 (Analysis) had the highest concerns, with low only in 4 studies (22.22%) and high in 14 (77.77%), due by limited validation, hyperparameter tuning, and reproducibility reporting. Overall quality concern was high in 15 studies (83.33%) and low in 3 (16.66%).

Assessment of applicability concern (PROBAST + AI)

Despite quality concerns, the applicability of AI models to the research question was generally good, with most studies showing low concern regarding the definition of predictors and outcomes, refer Fig. 5. However, He sarah et al. (2024) showed high concern, likely due to the use of synthetic data which may not perfectly translate to real biological samples. Applicability concerns are summarized in Fig. 6.

Domain 1 was low in 17 studies (94.44%) and high in 1 (5.55%). Domain 2 was low in 17 studies (94.44%) and high in 1 (5.55%). Domain 3 was low in 17 studies (94.44%) and high in 1 (5.55%). Overall applicability was



Fig. 3 PROBAST+AI quality concern ratings by domain and overall, for included eligible articles

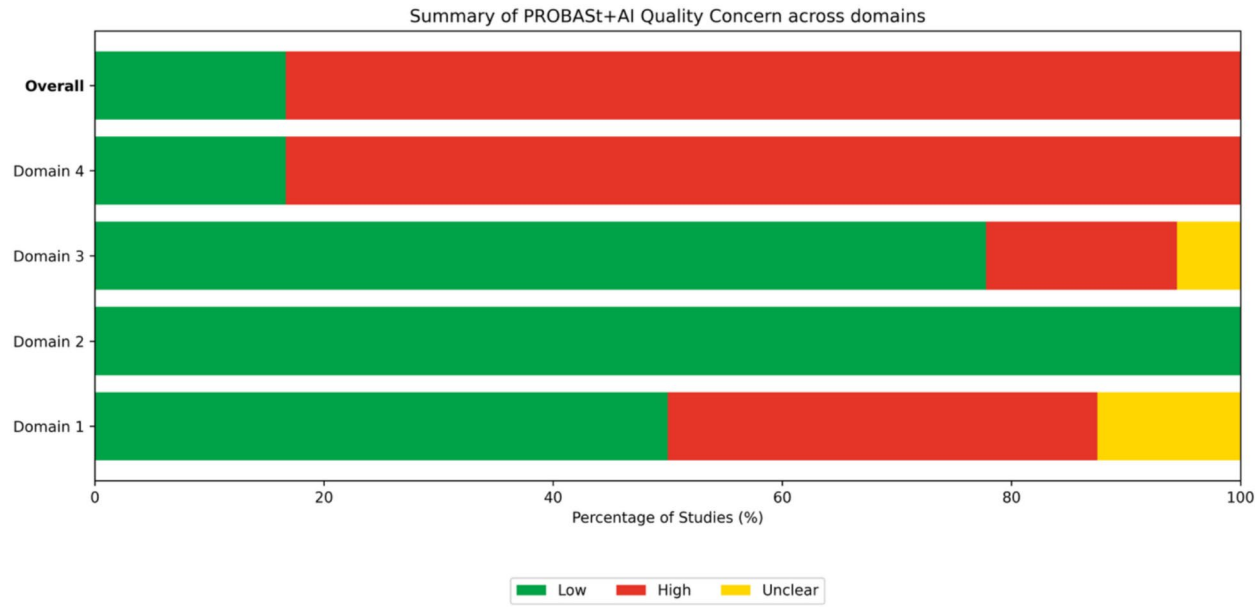


Fig. 4 Summary of PROBAST+AI quality concern across domain and overall, for included eligible articles

low in 17 studies (94.44%) and high in 1 (5.55%), indicating strong relevance to preclinical senescence research, with the single High rating due to synthetic data in one study (Fig. 2).

In contrast with the high quality concern, the assessment of applicability showed low concern in 94.44% of the studies (refer Fig. 5). This indicates a strong alignment between the AI model designs and the practical requirements of cellular senescence research. Specifically, in Domain 2 (Predictors), there was 100% low concern, reflecting that researchers are choosing biologically sound inputs such as nuclear morphology and established molecular markers that directly translate to standard laboratory assays. The only study flagged for high applicability concern (He, Sarah et al., 2024) was due to the heavy reliance on synthetic data. While synthetic data is useful for training, it may not capture the 'biological noise' and phenotypic heterogeneity found in real-world samples, potentially limiting its direct use in preclinical drug screening. Overall, these findings suggest that the field is building models that are highly relevant to the target biological questions, even if the underlying analytical rigor (as shown in the bias assessment) requires significant improvement for future translation.

Reporting TRIPOD + AI

The adherence to reporting standards was evaluated using the TRIPOD+AI checklist. The compliance rates for all 27 items are presented in Table 2. In Table 2, the abbreviations P, and N/A correspond to Partial and Not applicable The TRIPOD+AI assessment reflects significant gaps in open science and reproducibility. There is a significant lack of transparency in the field. 0% of studies pre-registered a protocol, which is a standard requirement for rigorous clinical research to prevent selective reporting. Furthermore, only 44.4% of studies shared their analytical code, making independent reproduction of the results difficult for the majority of the research. Average adherence for 'YES' was approximately 62%. Strong reporting was observed for Title, and objectives (100%), Outcome 8a (100%), Model development (100%), Limitations (100%), and Interpretation (100%). Gaps were evident in Sample Size Justification (33.33%), Class imbalance (50%), External Validation (30%), and Data Availability (72.22%), refer Fig. 6 for the visual percent 'YES' across all the items. By overview the recent studies and those with open repositories scored higher percentage of 'YES'.

The TRIPOD+AI assessment revealed that reporting quality was inconsistent across the included studies, with several recurring gaps that limited reproducibility and transparent appraisal of the models. Although many studies

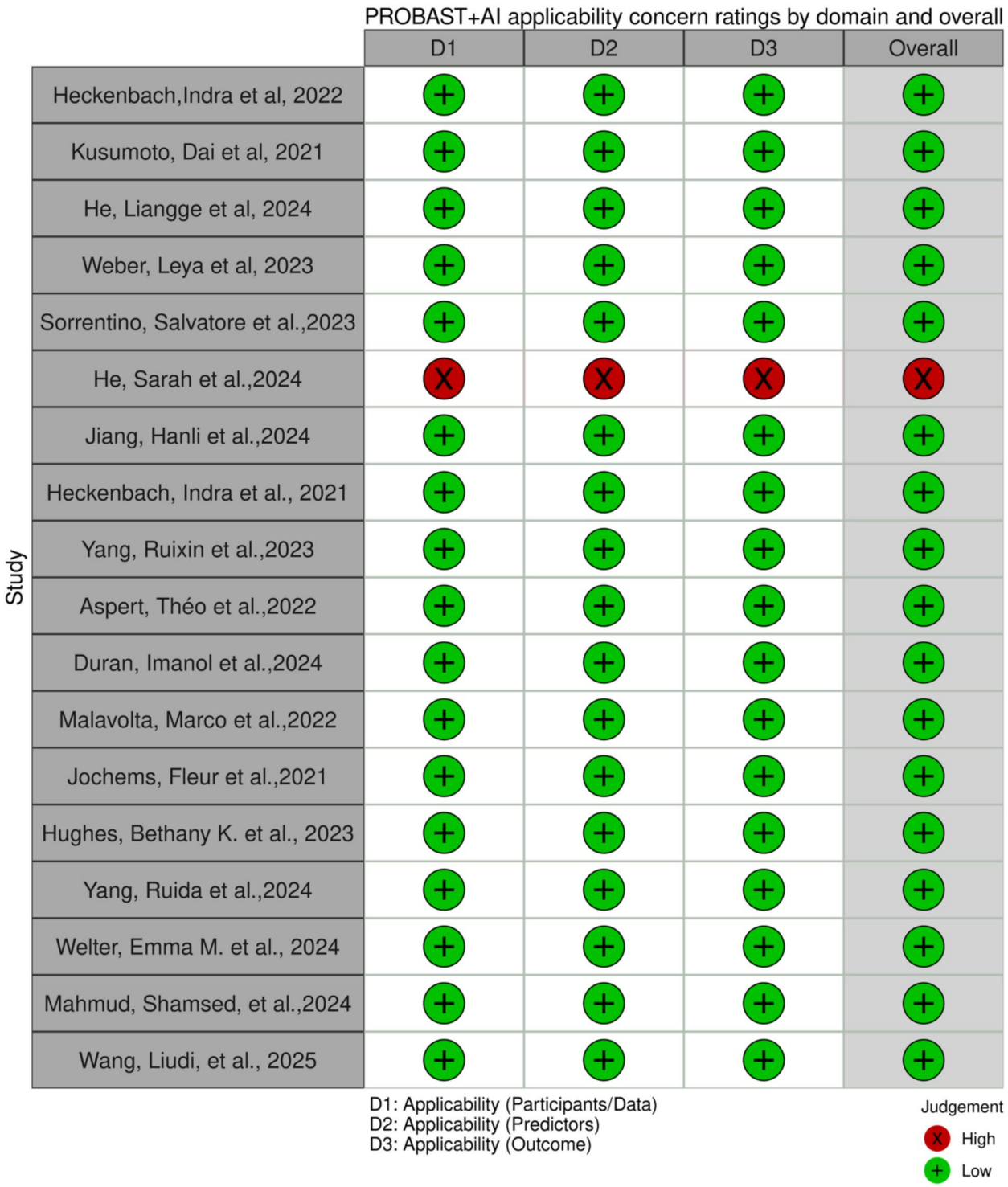


Fig. 5 PROBAST+AI applicability concern ratings by domain and overall, for included eligible articles

https://doi.org/10.1186/s12874-026-02912-2

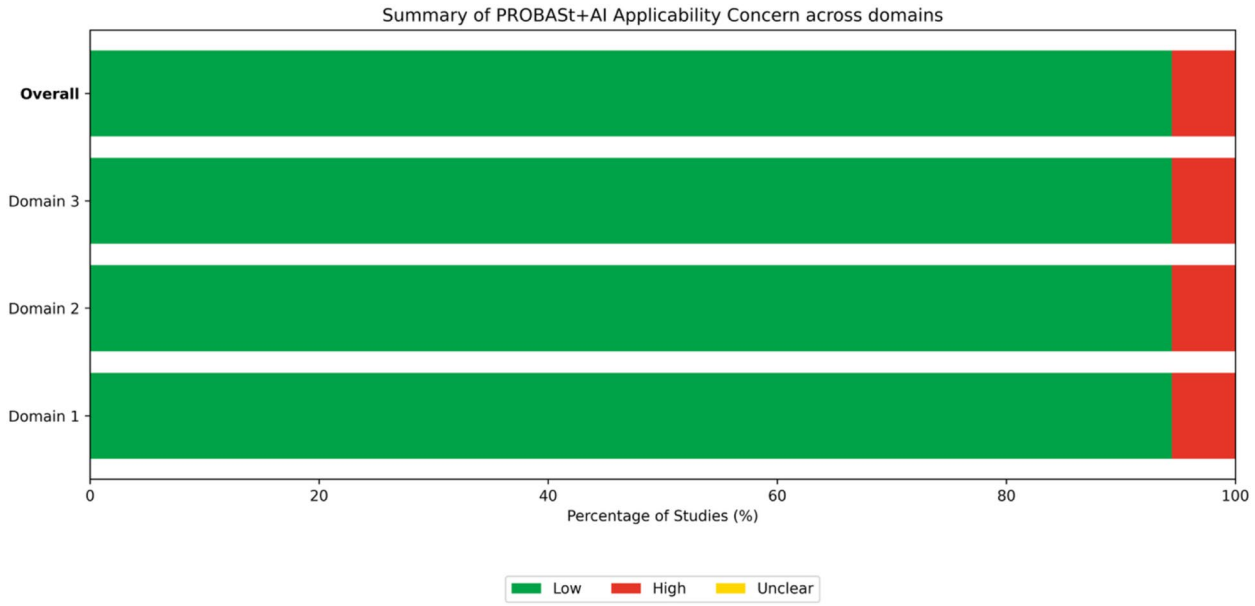
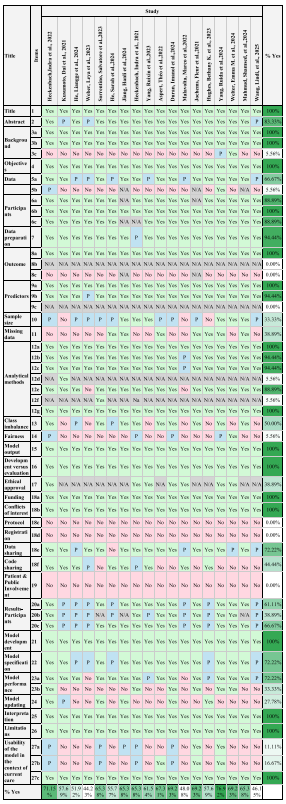


Fig. 6 Summary of PROBAST+AI applicability concern across domain and overall, for included eligible articles

Table 2 TRIPOD+AI reporting adherence across 27 key items for all 18 studies on AI-based senescence classification models [12, 18, 30–33, 35–46]



clearly described the AI task, data source, and headline performance metrics, fewer studies provided full reporting of protocol registration, code availability, and sample size justification, which are key elements for assessing the credibility and reproducibility of prediction model research.

A prominent deficiency was the near absence of protocol registration or pre-specified analysis plans. This omission makes it difficult to determine whether model development, feature selection, and performance evaluation were planned in advance or shaped after inspecting the data, increasing the possibility of selective reporting and analytic flexibility. Likewise, code sharing was limited in the majority of studies, reducing the ability of independent researchers to reproduce preprocessing steps, model architecture, parameter tuning, and final performance estimates.

Incomplete reporting of sample size rationale was another important limitation. Several studies reported model performance without explaining how the sample size was determined, whether the data were sufficient for the number of candidate predictors or model parameters, or how overfitting risk was controlled. This is especially important in AI-based senescence research, where datasets are often small, heterogeneous, and modality-specific, and where optimistic performance estimates may not generalize beyond the original study setting.

Overall, the TRIPOD + AI findings suggest that while the included studies demonstrate strong technical potential for senescence detection and classification, the reporting of development, validation, and transparency-related details remains incomplete. Strengthening protocol registration, code availability, and sample size reporting would substantially improve reproducibility and facilitate more reliable comparison across future AI studies in cellular senescence. This is especially important in AI-based senescence research, where datasets are often small and modality-specific. For example, while studies using high dimensional transcriptomics or specialized imaging flow cytometry showed high technical potential, the lack of sample size rationale makes it difficult to assess if these models were over-parameterized for their specific biological context.

Discussion

The integration of AI in cellular senescence research, as synthesized from 18 studies (refer Table 1), reveals several key trends. Like DL approaches, especially CNNs, transfer learning, and hybrids (dominant in ~ 67% of studies), excel at processing high-content microscopy images for morphological features (nuclear enlargement, granularity, shape changes), achieving accuracies often > 90%. ML methods (random forests, SVMs, ~ 30%) suit transcriptomic or proteomic data, while ensembles or transfer learning address small or noisy datasets. Biomarkers centre on morphology (38%), SA-β-gal or proteins (28%), and transcriptomics (17%), with promising multi-modal strategies (e.g., Wang et al., 2025 combining morphology + Ki67/p16/p21/p53/pRPS6). Outputs span classification (44%), scoring for drug efficacy/phenotyping (17%), detection, and prognostic modelling (e.g., cancer senescence burden).

Strengths include automation reducing subjectivity or bias, scalability for high-throughput screening, and predictive power for senescence trajectories, SASP and drug responses. Performance visualizations from Fig. 2a – f shows accuracy as the most reported metric (78%), followed by precision/recall/F1/AUC/mAP which outperforming traditional assays in speed or with the objectivity.

Critical evaluation of methodological risks (PROBAST + AI)

However, critical limitations are evident when evaluating these high performance metrics against our quality assessments. The application of the PROBAST + AI framework revealed a pervasive high risk of bias across 83.33% of the included studies, driven by distinct systemic vulnerabilities within specific domains.

In Domain 1 (Participant Selection and Data Sources), a high risk of bias was flagged in 33.33% of the literature, primarily due to a heavy reliance on small, unstandardized, or entirely proprietary datasets. When an AI model is trained on data unique to a single laboratory's setup, it introduces selection bias, which severely limits

external validity and prevents the computational features from easily generalizing to independent facilities operating under different culture conditions.

Conversely, Domain 2 (Predictors) demonstrated a 100% low risk of bias across all evaluated studies. Researchers consistently utilized scientifically justified, high-value input features deeply rooted in established senescence biology, such as cell and nuclear morphology, aspect ratios, convexity, and molecular hallmarks like p16, p21, and SA-β-gal.

Vulnerabilities in Domain 3 (Outcome Measurement) showed high or unclear concern in 16.66% of the studies, stemming directly from the limitations of ground truth verification. Because defining a definitive senescent state is complicated by phenotypic heterogeneity, studies relying on subjective, manual visual scoring of staining intensities as their absolute target outcome inadvertently introduced human cognitive bias into the model's training foundations.

The single largest driver of methodological weakness was Domain 4 (Analytical Methods), where 77.77% of the studies exhibited high concern due to improper data splitting and a total absence of external validation. The architectural consequence of splitting datasets at the cellular level rather than by independent biological replicate or patient source is severe data leakage. When cells from the exact same culture dish or batch are distributed across both training and validation sets, the model fails to learn generalizable biological features of senescence. Instead, it memorizes localized batch effects, such as minor illumination variations or background artifacts unique to that individual dish.

Downstream impact on validity and generalizability

This analytical flaw creates a striking "Performance-Validation Paradox" across the literature. While the included models report exceptionally high internal validation metrics with accuracies ranging from 83.55% to 99.79% these figures are heavily inflated by data leakage. Because over 70% of the models completely lacked external validation on entirely independent datasets, their true generalizability remains unverified. If these models are transitioned from controlled preclinical laboratory settings to real-world applications, such as assessing in vivo tissue burden or guiding clinical trials, they are highly prone to performance collapse due to overfitting on their original training batches.

Despite these severe internal quality concerns, the applicability of the models to the core research questions was remarkably strong, with 94.44% showing low concern across all domains. This confirms that the field is successfully targeting highly relevant biological outcomes and choosing appropriate predictors, even if the underlying mathematical execution requires a massive shift in analytical rigor to achieve clinical translation.

TRIPOD+AI adherence (~62% average) highlights transparency deficits with protocol registration/pre-registration (0%), low code sharing (44%), data availability gaps, and limited external validation/sample justification, refer Fig. 7 and Table 2. The pronounced heterogeneity across the datasets utilized, dominated by microscopy data (67%) and a scarcity of public, multi-omics, or clinical sources, compromises the ability to compare and generalize findings. This issue is further compounded by a lack of standardization in reported metrics and data preprocessing techniques. Given that most studies are currently confined to preclinical stages, the translation into real-world applications, such as assessing in vivo burden or conducting clinical trials, remains notably constrained.

The convergence of these findings with established challenges in biomedical AI, such as batch effects and issues of interpretability, highlights a specific set of difficulties within senescence research, notably phenotypic heterogeneity and the inherent variability of the Senescence-Associated Secretory Phenotype (SASP). While AI demonstrably accelerates biomarker discovery for instance, through the refinement of gene sets using techniques like T-CLASS and expedites senolytic compound screening compared to conventional methods, the absence of standardised protocols continues to impede reliable reproducibility. The profound implications of this work encompass faster therapeutic development for age-related disorders, the creation of superior prognostic models in oncology and regenerative medicine, and the fostering of interdisciplinary breakthroughs at the intersection of biology and artificial intelligence.

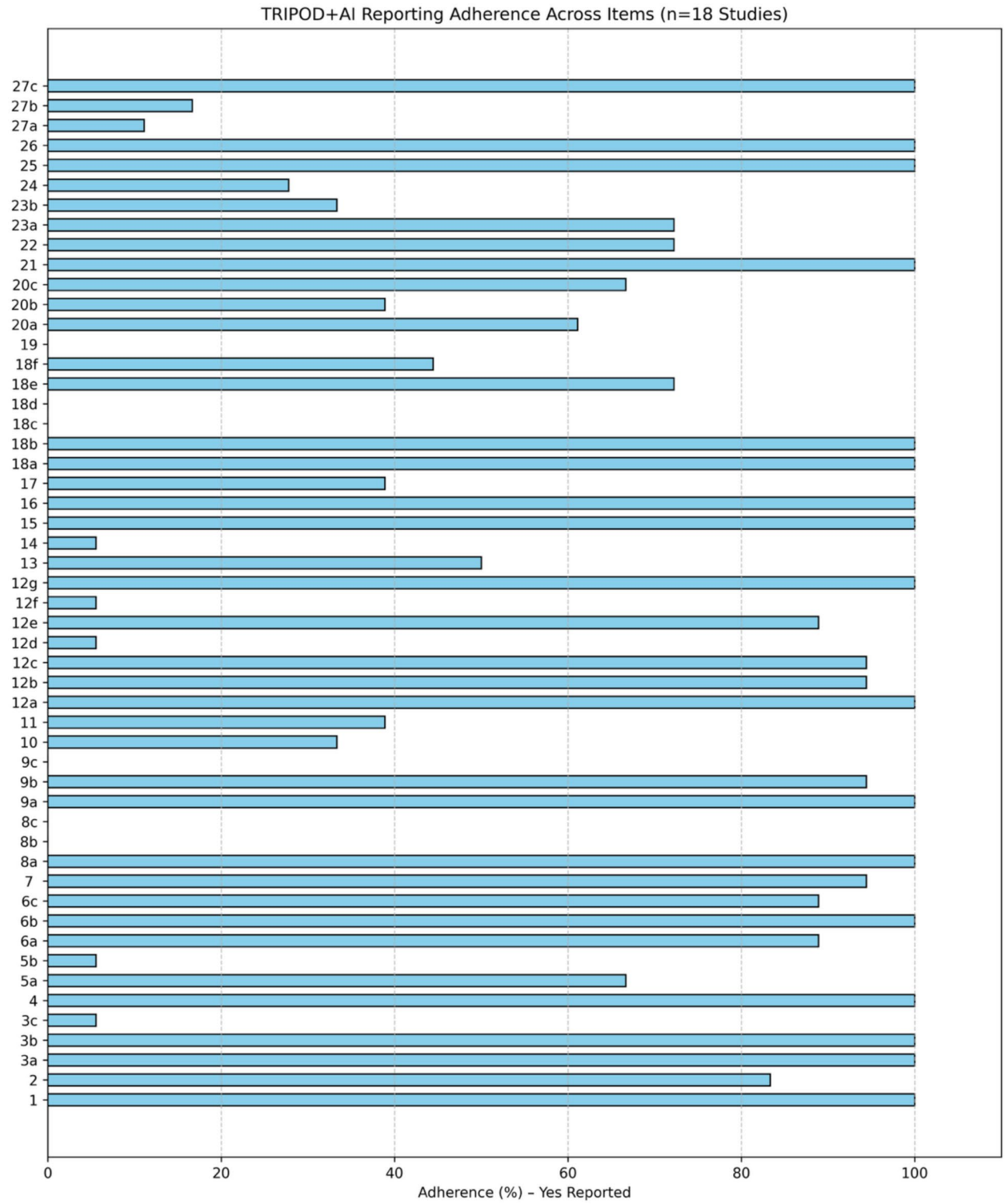


Fig. 7 TRIPOD+AI reporting adherence across all items from checklist

Methodological recommendations and best practices for future Bio-AI research

To address the pervasive quality and reporting deficits identified in this audit, future implementations of artificial intelligence in cellular senescence research must transition from a "performance-first" paradigm to a "protocol-first" framework. Based on the systemic gaps uncovered via the PROBAST + AI and TRIPOD + AI assessments, we propose the following concrete best practices for model development, validation, and transparent reporting:

Standardized data splitting and batch-effect mitigation

Researchers must abandon cell-level data splitting when handling microscopy datasets. Data partitioning into training, testing, and validation sets must occur strictly at the patient level, biological replicate level, or distinct cell-culture dish level. This architectural barrier is essential to prevent data leakage and ensure that the AI algorithm learns true biological features of senescence rather than local laboratory artifacts or illumination variables.

Mandatory independent external validation

Reporting high internal validation metrics (e.g., k-fold cross-validation) is insufficient to establish model generalizability. Future studies must test their finalized architectures on completely independent external datasets, ideally sourced from different laboratories, alternative cell lines, or varying senescence-induction methods (e.g., comparing hydrogen peroxide stress-induced senescence with replicative senescence).

Comprehensive sample size rationale and class imbalance handling

Authors must explicitly justify their sample size using established power analysis methods adapted for deep learning or regression frameworks to ensure models are not overparameterized. As well, when working with naturally imbalanced biological datasets (e.g., a low percentage of senescent cells in a healthy culture), studies must document their data-augmentation or loss-function weighting strategies to address class imbalance transparently.

Strict compliance with TRIPOD + AI and open science standards

To bridge the current reproducibility deficit, journal editors and reviewers should mandate full compliance with the 2024/2025 TRIPOD + AI reporting checklist during manuscript submission. Crucially, future bio-AI investigators must abandon ad-hoc, selective performance reporting and mandate the uniform co-reporting of robust metrics specifically Area Under the ROC Curve (AUC), F1-scores, sensitivity, and specificity. Moving away from an isolated reliance on raw accuracy is vital for managing biological class imbalances and allowing for reliable, cross-study comparisons. Additionally, future studies must prioritize prospective protocol registration (e.g., on platforms like OSF or ClinicalTrials.gov) to eliminate analytic flexibility, while making all data preprocessing pipelines, custom model weights, and analytical code publicly available in open repositories like GitHub or Zenodo.

Conclusion

This PRISMA guided systematic review of 18 studies demonstrates that AI in particular DL on morphological data provides high accuracy (typically 85 to 99%), automated tools for senescent cell detection, classification, scoring and prognostic modelling, addressing key limitations of conventional essays. Common applications leverage nuclear or cellular morphology, SA-β-gal and molecular markers with hybrid or transfer learning improving robustness on diverse datasets. However, prevasive high risks of bias (PROBAST + AI, mainly analysis domain)

and suboptimal reporting (TRIPOD+AI adherence approximately 62%, with critical gaps in code or protocol sharing and validation) temple confidences current models. Overall, AI holds transformative potential for senescence biology and ageing interventions, but rigorous standardisation is required to realise reliable reproducible and clinically applicable tools.

Future research endeavours should prioritise the establishment of shared, meticulously annotated benchmark datasets. These datasets must encompass diverse cell types, various senescence-inducing stimuli (including replicative, stress-induced, and oncogene-driven senescence), and multiple data modalities (such as high-resolution microscopy integrated with multi-omics data and clinical information). This foundational step is crucial for enabling rigorous, direct comparisons between computational models and mitigating the current heterogeneity across studies. The standardization of preprocessing pipelines, evaluation metrics (e.g., the consistent reporting of F1-scores or AUC with appropriate confidence intervals), and research reporting (adhering strictly to the full TRIPOD+AI guidelines, including transparent code and data availability in public repositories) is paramount for bolstering reproducibility and fostering global scientific collaboration.

The integration of multi-omics data streams specifically transcriptomics, proteomics, and epigenomics, through advanced multimodal transformer architectures holds the potential to construct comprehensive, holistic profiles of cellular senescence, thereby uncovering novel diagnostic biomarkers and disease trajectories. This evolution directly mirrors broader advancements in medical image analysis, where the transition from classic CNNs to recurrent 3-D multimodal transformers and composite image fusion has significantly enhanced diagnostic accuracy and early disease detection [48–50]. A concentrated focus on explainable AI (XAI) techniques, such as feature importance mapping, attention mechanisms, and SHAP/LIME methodologies, is essential to enhance model interpretability. Drawing inspiration from recent advancements in digital histopathology, the implementation of multi-channel attention mechanisms and interpretable transfer learning frameworks such as those successfully utilized for gastric cancer classification can provide transparent, quantitative insights into morphological changes [51, 52]. This interpretability is not merely a technical requirement but a critical prerequisite for meaningful biological validation and eventual regulatory acceptance in clinical trial settings.

As well the implementation of longitudinal studies and real-time monitoring strategies (e.g., combining time-lapse imaging with AI) in conjunction with senolytic or other clinical interventions will allow for the dynamic tracking of senescence responses. This evidence will directly inform precision medicine approaches in the context of aging and associated pathologies, including cancer, neurodegeneration, and fibrosis. Broader external validation across independent laboratories and model systems, alongside rigorous assessments of fairness and potential bias (considering demographic and cell-type diversity), and the expansion of large-scale public datasets (e.g., generating TCGA-like resources specifically for senescence research) are imperative for strengthening generalizability. Ultimately, the development of harmonized, end-to-end AI workflows from initial data curation to final model deployment, will significantly accelerate the pace of discovery, facilitate the formation of international consortia, and solidify the role of artificial intelligence as a cornerstone technology for the development of healthy longevity interventions.

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1186/s12874-026-02912-2.

Acknowledgements Not applicable.

Authors' contributions CR did PROBAST+AI and TRIPOD+AI analysis. BR performed the systematic review using PRISMA guidelines. GP Initiated the core idea for the study, provided critical intellectual input and guidance, and contributed to manuscript review and refinement. All authors read and approved the final manuscript.

Funding This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Data availability No datasets were generated or analysed during the current study.

Declarations

Ethics approval and consent to participate Not applicable.

Consent for publication Not applicable.

Competing interests The authors declare no competing interests.

Open Access This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by-nc-nd/4.0/.

References

1. Shreeya T, Ansari MS, Kumar P, Saifi M, Shati AA, Alfaifi MY, et al. Senescence: a DNA damage response and its role in aging and neurodegenerative diseases. Front Aging. 2024;4:1292053.
2. Ramesh T, Singh R, Barage SH. Epigenetics of aging and epigenetic modifications of aging-associated diseases. In: Drug discovery and antiaging approaches for human longevity. IGI Global Scientific Publishing; 2025. p. 239–82.
3. Di Micco R, Krizhanovsky V, Baker D, d'Adda di Fagagna F. Cellular senescence in ageing: from mechanisms to therapeutic opportunities. Nat Rev Mol Cell Biol. 2021;22(2):75–95.
4. Lyu YX, Fu Q, Wilczok D, Ying K, King A, Antebi A, ... & Mansukhani, R. Longevity biotechnology: bridging AI, biomarkers, geroscience and clinical applications for healthy longevity. Aging (Albany NY) 2024; 16(20): 12955.
5. Lushchak O, Schosserer M, Grillari J. Senopathies—diseases associated with cellular senescence. Biomolecules. 2023;13(6):966.
6. Sacco A, Belloni L, Latella L. From development to aging: the path to cellular senescence. Antioxid Redox Signal. 2021;34(4):294–307.
7. Rossi M, Abdelmohsen K. The emergence of senescent surface biomarkers as senotherapeutic targets. Cells. 2021;10(7):1740.
8. Lucas V, Cavadas C, Aveleira CA. Cellular senescence: from mechanisms to current biomarkers and senotherapies. Pharmacol Rev. 2023;75(4):675–713.
9. Zubair M, Umair M, Naqvi RA, Hussain D, Owais M, Werghi N. A comprehensive computer-aided system for an early-stage diagnosis and classification of diabetic macular edema. J King Saud Univ Comput Inform Sci. 2023;35(8):101719.
10. Zubair M, Umair M & Owais M. Automated brain tumor detection using soft computing-based segmentation technique. In 2023 3rd International Conference on Computing and Information Technology (ICCIT) (pp. 211–215). IEEE. 2023.
11. Krentzel D, Shorte SL, Zimmer C. Deep learning in image-based phenotypic drug discovery. Trends Cell Biol. 2023;33(7):538–54.
12. Welter EM, Benavides S, Archer TK, Kosyk O, Zannas AS. Machine learning-based morphological quantification of replicative senescence in human fibroblasts. Geroscience. 2024;46(2):2425–39.
13. Steurer B, Vanhaelen Q, Zhavoronkov A. Multimodal transformers and their applications in drug target discovery for aging and age-related diseases. J Gerontol A Biol Sci Med Sci. 2024;79(9):glae006.
14. Farrell S, Mitnitski A, Rockwood K, Rutenberg AD. Interpretable machine learning for high-dimensional trajectories of aging health. PLoS Comput Biol. 2022;18(1):e1009746.
15. Mai M, Luo S, Fasciano S, Oluwole TE, Ortiz J, Pang Y, et al. Morphology-based deep learning approach for predicting adipogenic and osteogenic differentiation of human mesenchymal stem cells (hMSCs). Front Cell Dev Biol. 2023;11:1329840.
16. Nagao Y, Sakamoto M, Chinen T, Okada Y, Takao D. Robust classification of cell cycle phase and biological feature extraction by image-based deep learning. Mol Biol Cell. 2020;31(13):1346–54.
17. Lee SCJ, Lee GY, Kim SS, Bae Y, Ham S, Sohn J, et al. T-CLASS: an online tool for the identification and classification of aging and senescence using transcriptome data. Aging Cell. 2025. https://doi.org/10.1111/acel.70193.

18. Mahmud S, Zheng C, Santiago FE, Zhang L, Robbins PD, Dong X. A machine learning approach identifies cellular senescence on transcriptome data of human cells in vitro. Geroscience. 2024. https://doi.org/10.1007/s11357-024-01485-6.
19. Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, ... & Moher D. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ. 2021; 372.
20. Lu YA, Lin HC, Tsai PS. Effects of digital sleep interventions on sleep among college students and young adults: systematic review and meta-analysis. J Med Internet Res. 2025;27:e69657.
21. Moher D, Liberati A, Tetzlaff J, Altman DG, Prisma Group. Preferred reporting items for systematic reviews and meta-analyses: the PRISMA statement. Int J Surg. 2010;8(5):336–41.
22. Haddaway NR, Page MJ, Pritchard CC, McGuinness LA. PRISMA2020 : an R package and Shiny app for producing PRISMA 2020-compliant flow diagrams, with interactivity for optimised digital transparency and Open Synthesis. Campbell Syst Rev. 2022;18(2):e1230. https://doi.org/10.1002/cl2.1230.
23. Faizal MS. B. M., & Osman KB. Exploring Effective Storytelling Modalities among Teenagers in Learning Science: A Systematic.
24. Lin S, Evans K, Hartley D, Morrison S, McDonald S, Veidt M, et al. A review of gait analysis using gyroscopes and inertial measurement units. Sensors (Basel). 2025;25(11):3481.
25. Eini P, Rezayee M, Tremblay J. Automated assessment of right heart function by artificial intelligence: a systematic review and meta-analysis. Eur J Radiol Open. 2026;16:100713.
26. McGuinness LA, Higgins JPT. Risk-of-bias VISualization (robvis): an R package and Shiny web app for visualizing risk-of-bias assessments. Res Synth Methods. 2020(1). https://doi.org/10.1002/jrsm.1411.
27. Papadopoulou E, Sapalidis K, Triaridis S, Printza A. The role of primary repair of the recurrent laryngeal nerve during thyroid/parathyroid surgery in vocal outcomes—a systematic review. J Clin Med. 2023;12(3):1212.
28. Collins GS, Moons KG, Dhiman P, Riley RD, Beam AL, Van Calster B, ... & Logullo P. TRIPOD+ AI statement: updated guidance for reporting clinical prediction models that use regression or machine learning methods. BMJ. 2024; 385.
29. Park WD, DeLonais-Parker A, Okeke B, Lentine KL, Benjamin MM, SLU Transplant Rx research Group. Machine learning for predicting long-term cardiovascular outcomes in kidney transplant recipients. JACC Adv. 2025;4(12):102364.
30. Kusumoto D, Seki T, Sawada H, Kunitomi A, Katsuki T, Kimura M, ... & Yuasa S. (2021). Anti-senescent drug screening by deep learning-based morphology senescence scoring. Nature Commun 2021; 12(1):257.
31. Jiang H, Lin L, Deng D, Ren J, Yang X, Liu S, et al. SenSeqNet: a deep learning framework for cellular senescence detection from protein sequences. bioRxiv. 2024;2024:10.
32. He S, Sillah M, Cole AR, Uboveja A, Aird KM, Chen YC, et al. D-MAINS: a deep-learning model for the label-free detection of mitosis, apoptosis, interphase, necrosis, and senescence in cancer cells. Cells. 2024;13(12):1004.
33. Yang R, Du Y, Kwan W, Yan R, Shi Q, Zang L, et al. A quick and reliable image-based AI algorithm for evaluating cellular senescence of gastric organoids. Cancer Biol Med. 2023;20(7):1.
34. Moons KG, Damen JA, Kaul T, Hooft L, Navarro CA, Dhiman P, ... & van Smeden M. PROBAST+ AI: an updated quality, risk of bias, and applicability assessment tool for prediction models using regression or artificial intelligence methods. BMJ. 2025; 388.
35. Heckenbach I, Mkrtchyan GV, Ezra MB, Bakula D, Madsen JS, Nielsen MH, et al. Nuclear morphology is a deep learning biomarker of cellular senescence. Nat Aging. 2022;2(8):742–55.
36. He L, Li M, Wang X, Wu X, Yue G, Wang T, ... & Zhou G. Morphology-based deep learning enables accurate detection of senescence in mesenchymal stem cell cultures. BMC Biol. 2024; 22(1):1.
37. Weber L, Lee BS, Imboden S, Hsieh CJ, Lin NY. Phenotyping senescent mesenchymal stromal cells using AI image translation. Curr Res Biotechnol. 2023;5:100120.
38. Sorrentino S, Manetti F, Bresci A, Vernuccio F, Ceconello C, Ghislanzoni S, ... & Polli D. (2023). Deep ensemble learning and transfer learning methods for classification of senescent cells from nonlinear optical microscopy images. Front Chem. 2023; 11: 1213981.
39. Heckenbach I, Ezra MB, Mkrtchyan GV, Madsen JS, Nielsen MH, Oró D, ... & Scheibye-Knudsen M. Deep learning shows cellular senescence is a barrier to cancer development. bioRxiv. 2021; 2021–03.
40. Aspert T, Hentsch D, Charvin G. DetecDiv, a generalist deep-learning platform for automated cell division tracking and survival analysis. Elife. 2022;11:e79519.
41. Duran I, Pombo J, Sun B, Gallage S, Kudo H, McHugh D, ... & Gil J. Detection of senescence using machine learning algorithms based on nuclear features. Nature Commun. 2024; 15(1):1041.
42. Malavolta M, Giacconi R, Piacenza F, Strizzi S, Cardelli M, Bigossi G, ... & Provinciali M. Simple detection of unstained live senescent cells with imaging flow cytometry. Cells. 2022; 11(16): 2506.
43. Jochems F, Thijssen B, De Conti G, Jansen R, Pogacar Z, Groot K, et al. The cancer SENESCopedia: a delineation of cancer cell senescence. Cell Rep. 2021. https://doi.org/10.1016/j.celrep.2021.109441.
44. Hughes BK, Davis A, Milligan D, Wallis R, Philpott MP, Wainwright LJ, ... & Bishop CL. SenPred: A single-cell RNA sequencing-based machine learning pipeline to classify senescent cells for the detection of an in vivo senescent cell burden. bioRxiv. 2023; 2023–10.

45. Yang R, Sun F, Shi Y, Wang H, Fan Y, Wu Y, et al. Machine learning-based development and validation of a cell senescence predictive and prognostic signature in intrahepatic cholangiocarcinoma. J Cancer. 2024;15(9):2810.
46. Wang L, Zhang H, Chen J, Xue X, Chen Y, Gao T, et al. Senescence screening of mesenchymal stromal cells by morphology-based deep learning senescence analysis. Biomed Signal Process Control. 2025;104:107549.
47. Brullas G, Luaces C, Trenchs V, Brotons P. Machine learning models for predicting hospital admission in pediatric emergency departments: a systematic review. Int J Med Inform. 2025. https://doi.org/10.1016/j.ijmedinf.2025.106189.
48. Zubair M, Ahmad PN, Anwar MS, Ahmad M & Masood S. DECODE-PD: deep explainable composite imaging fusion of MRI, PET, and SPECT for early detection of Parkinson's disease. In Medical Imaging 2026: Computer-Aided Diagnosis (Vol. 13926, pp. 600–603). SPIE. 2026.
49. Owais M, Zubair M, Hassan T, Velayudhan D, Hussain I & Werghi N. Recurrent 3-D multi-level visual transformer for joint classification of heterogeneous 2-d AND 3-D radiographic data. In 2024 IEEE international conference on image processing (ICIP) (pp. 3205–3211). IEEE. 2024.
50. Owais M, Zubair M, Kareem DNA, Naseer SA, Alam M, Ahmad M & Hussain I. A comprehensive review tracing the evolution of volumetric medical imaging analysis from classic CNNs to emerging AI-agents. 2025.
51. Zubair M, Owais M, Hassan T, Bendechache M, Hussain M, Hussain I, et al. An interpretable framework for gastric cancer classification using multi-channel attention mechanisms and transfer learning approach on histopathology images. Sci Rep. 2025;15(1):13087.
52. Zubair M, Owais M, Mahmood T, Iqbal S, Usman SM, Hussain I. Enhanced gastric cancer classification and quantification interpretable framework using digital histopathology images. Sci Rep. 2024;14(1):22533.

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Authors and Affiliations

Chanda Bhalchandra Rajurkar[1] · Bala Murugan[1] · Ganesh Pandian Namasivayam[2]

✉ Bala Murugan
balamurugan.ms@vit.ac.in

[1] School of Electronics Engineering, Vellore Institute of Technology, Chennai, India

[2] Institute for Integrated Cell-Material Sciences (iCeMS), Kyoto University, Kyoto, Japan